Son of Surya

GUIDE
R11
FUNDAMENTALS

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Solar Dictionary English–Hindi

About 150 solar, electricity, billing, scheme and farm-solar terms — each with the Hindi word you will actually hear, and a plain-language meaning. सौर शब्दकोश

kWh	Wp	DC	Net
यूनिट	वॉट-पीक	दिष्ट धारा	नेट मीटरिंग

“Understand your solar before you buy your solar.”

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R11 About This Dictionary

Guide	**R11 — Solar Dictionary (English–Hindi)** · सौर शब्दकोश (अंग्रेज़ी–हिंदी)
Category · Level · Price	Solar Fundamentals · All levels · **Free**
For	Homeowners, farmers, students, installers and anyone reading solar quotes, bills, scheme forms or datasheets
Use with	Every Son of Surya guide — each defines its own terms, and this dictionary collects them in one place

The problem

Solar is explained in English technical words, but many Indian buyers, farmers and family decision-makers think and talk in Hindi or a regional language. Misunderstood words become wrong decisions.

The solution

About 150 terms, each with the English word, the common Hindi word (often the English word written in Devanagari, as it is actually spoken), and a plain-language meaning.

How to use this dictionary

- Terms are grouped by topic, from basic electricity to farm solar, then abbreviations and a unit cheat sheet.
- The Hindi column gives the word you will most likely **hear or read** in India. Where a pure Hindi word exists but is rarely used in practice, both are given.
- Many technical words — इन्वर्टर, पैनल, मीटर — are used in English form even in Hindi. That is normal, not a mistake.
- Meanings are simplified for learning. Technical standards and scheme documents give the formal definitions.

WORD PAIRS PEOPLE MIX UP

kW (power — how fast)	vs	**kWh / unit** (energy — how much)
Irradiance (W/m², strength now)	vs	**Irradiation** (kWh/m², total over time)
Solar inverter (solar DC → AC)	vs	**Home UPS "inverter"** (battery backup)
Wp (lab rating of a panel)	vs	**W** (actual power at a moment)
Ah (battery charge)	vs	**Wh / kWh** (battery energy = V × Ah)
Sanctioned load (kW limit)	vs	**Consumption** (kWh used)
Product warranty (defects)	vs	**Performance warranty** (output over years)

01 Electricity Basics · बिजली की बुनियादी बातें

English	हिंदी	Simple meaning
Electricity	बिजली / विद्युत	Flow of electric charge that powers appliances
Voltage (V)	वोल्टेज / विभवांतर	The “push” of electricity, in volts
Current (A)	करंट / विद्युत धारा	Amount of electricity flowing, in amperes
Resistance (Ω)	प्रतिरोध	Opposition to flow; causes heating
Power (W, kW)	पावर / शक्ति	How fast energy is used or produced
Energy (kWh)	ऊर्जा	Total electricity used or produced over time
Unit	यूनिट / इकाई	1 kWh — what your bill counts
Watt	वॉट	Unit of power
Kilowatt (kW)	किलोवाट	1,000 watts
Kilowatt-hour (kWh)	किलोवाट-घंटा	1,000 watts for one hour = 1 unit
Alternating current (AC)	प्रत्यावर्ती धारा (AC)	Current that reverses direction; grid supply
Direct current (DC)	दिष्ट धारा (DC)	One-direction current; from panels and batteries
Frequency (Hz)	आवृत्ति (हर्ट्ज़)	AC cycles per second; India 50 Hz
Phase (single / three)	फेज़ (सिंगल / थ्री)	Type of supply connection
Power factor	पावर फैक्टर / शक्ति गुणांक	Share of AC current doing useful work (0–1)
Apparent power (VA, kVA)	आभासी शक्ति (VA)	V × I in AC, before power factor
Load	लोड / भार	Appliances drawing power, or total power drawn
Short circuit	शॉर्ट सर्किट	Accidental low-resistance path; dangerous current
Efficiency	दक्षता	Useful output ÷ input
Horsepower (HP)	हॉर्सपावर / अश्वशक्ति	Pump/motor rating; 1 HP ≈ 0.746 kW

02 Sun & Site · सूर्य और स्थल

English	हिंदी	Simple meaning
Solar energy	सौर ऊर्जा	Energy from sunlight
Irradiance	सौर विकिरण तीव्रता	Strength of sunlight now, W/m²
Irradiation (insolation)	सौर विकिरण (कुल)	Sunlight energy over a day/year, kWh/m²
Peak sun hours	पीक सन आवर्स	Daily sunlight expressed as hours at 1,000 W/m²

GHI	क्षैतिज सौर विकिरण	Sunlight on a flat surface
Tilt angle	झुकाव कोण	Panel angle from horizontal
Orientation / azimuth	दिशा / दिगंश	Direction panels face (south in India)
Shade / shadow	छाया	Blocked sunlight; reduces output
Shade-free area	छाया-मुक्त क्षेत्र	Roof area with no shadow in productive hours
Sun path	सूर्य पथ	Sun's track across the sky through the year
Solar elevation	सूर्य की ऊँचाई (उन्नयन कोण)	Sun's angle above the horizon
Soiling	धूल-गंदगी जमना	Dust and dirt on panels
Monsoon	मानसून	Rainy season; lower solar output
Rooftop	छत (रूफटॉप)	Roof used for panels
Ground-mounted	ज़मीन पर लगा (ग्राउंड-माउंटेड)	Panels installed on land
Specific yield	प्रति kWp उत्पादन	Units per kWp per day or year

03 Equipment · उपकरण

English	हिंदी	Simple meaning
Solar cell	सौर सेल	Smallest unit that turns light into electricity
Solar panel / module	सोलर पैनल / सौर मॉड्यूल	Many cells joined in one frame
Watt-peak (Wp)	वॉट-पीक	Panel rating under standard test conditions
Array	पैनल समूह (ऐरे)	All panels of a system
String	स्ट्रिंग	Panels connected in series
Inverter	इन्वर्टर	Converts DC to AC
Grid-tied inverter	ग्रिड-टाई इन्वर्टर	Inverter that feeds home and grid; no backup
Hybrid inverter	हाइब्रिड इन्वर्टर	Manages solar, battery and grid
Home UPS	घरेलू इन्वर्टर / यूपीएस	Battery backup unit charged from the grid
Micro-inverter / optimiser	माइक्रो-इन्वर्टर / ऑप्टिमाइज़र	Panel-level electronics; reduce shade losses
Charge controller	चार्ज कंट्रोलर	Controls battery charging from panels
MPPT	एमपीपीटी	Tracks panels' best operating point
Battery	बैटरी	Stores energy
Lithium (LFP) battery	लिथियम (LFP) बैटरी	Common lithium chemistry; deeper usable share
Lead-acid battery	लेड-एसिड बैटरी	Traditional battery; lower usable share

Ampere-hour (Ah)	एम्पियर-घंटा	Battery charge; × V = Wh
Depth of discharge	डिस्चार्ज की गहराई (DoD)	Share of battery capacity used
Mounting structure	माउंटिंग ढाँचा / स्ट्रक्चर	Frame holding panels
Junction box	जंक्शन बॉक्स	Box where cables join
ACDB / DCDB	एसी / डीसी वितरण बॉक्स	Boxes with protection on AC and DC sides
Cable	केबल / तार	Conductor carrying electricity
Fuse	फ्यूज़	Melts to stop excess current
Circuit breaker (MCB)	सर्किट ब्रेकर (एमसीबी)	Switch that trips on overload
RCD / RCCB	आरसीसीबी (रिसाव सुरक्षा)	Trips on leakage current; protects people
Surge protection device	सर्ज सुरक्षा उपकरण (SPD)	Protects from voltage spikes
Lightning arrester	तड़ित चालक / लाइटनिंग अरेस्टर	Gives lightning a safe path to earth
Earthing	अर्थिंग / भू-संपर्कन	Safe connection of metal parts to ground
Earth pit	अर्थिंग पिट	Buried electrode for earthing
Bypass diode	बाईपास डायोड	Routes current around shaded cells
Datasheet	डेटाशीट / तकनीकी विवरण पत्र	Manufacturer's specification sheet

04 System Types & Design · सिस्टम के प्रकार

English	हिंदी	Simple meaning
On-grid system	ऑन-ग्रिड सिस्टम	Connected to the grid, usually no battery
Off-grid system	ऑफ-ग्रिड सिस्टम	Not connected to the grid; uses batteries
Hybrid system	हाइब्रिड सिस्टम	Grid + battery + solar
Anti-islanding	एंटी-आइलैंडिंग	Inverter shuts off when grid fails, for safety
System size (kWp)	सिस्टम क्षमता (kWp)	Total panel rating
Standard Test Conditions	मानक परीक्षण स्थितियाँ (STC)	1,000 W/m², 25 °C, AM 1.5 lab conditions
Temperature coefficient	तापमान गुणांक	How much output falls per °C of heating
Performance ratio (PR)	परफॉर्मेंस रेशियो	Actual ÷ ideal output; overall losses
Capacity utilisation factor (CUF)	क्षमता उपयोग कारक	Yearly output ÷ (kWp × 8,760 h)
Degradation	क्षरण / गिरावट	Slow loss of panel output over years
Losses	हानियाँ (लॉसेज़)	Energy lost as heat, shade, dust, conversion

Single-line diagram	सिंगल लाइन डायग्राम	Simplified electrical drawing of the system
Layout drawing	लेआउट ड्रॉइंग	Plan of panel positions on the roof
Wind load	पवन भार	Force of wind on panels and structure
Corrosion	जंग / संक्षारण	Rusting of metal, worse near the sea
Agrivoltaics	कृषि-सौर (एग्रीवोल्टाइक्स)	Crops and solar on the same land

05 Grid, Bills & Metering · ग्रिड, बिल और मीटरिंग

English	हिंदी	Simple meaning
Grid	ग्रिड / विद्युत जाल	The public electricity network
DISCOM	बिजली वितरण कंपनी (डिस्कॉम)	Company that supplies and bills you
Consumer number	उपभोक्ता क्रमांक	Your account number with the DISCOM
Consumer category	उपभोक्ता श्रेणी	Residential, commercial, agricultural etc.
Sanctioned load	स्वीकृत भार	Maximum power your connection is approved for
Tariff	टैरिफ / बिजली दर	Price of electricity
Slab	स्लैब	Block of units charged at one rate
Fixed charge	स्थिर शुल्क (फिक्स्ड चार्ज)	Monthly charge for the connection
Energy charge	ऊर्जा शुल्क	₹ per unit consumed
Electricity duty	विद्युत शुल्क (ड्यूटी)	State tax on electricity
Fuel adjustment charge	ईंधन समायोजन शुल्क (FAC)	Variable charge for fuel/power-purchase cost
Meter reading	मीटर रीडिंग	Number shown by the meter
Multiplying factor	गुणांक (MF)	Number to multiply meter difference by
Net metering	नेट मीटरिंग	Export set against import in billing
Gross metering	ग्रॉस मीटरिंग	All solar output sold to the grid at a set rate
Bi-directional meter	द्वि-दिशात्मक मीटर	Meter counting import and export
Import / export	आयात / निर्यात (बिजली)	Electricity from / to the grid
Banking (units)	यूनिट बैंकिंग	Surplus units carried forward, under state rules
Time-of-day tariff	समय-आधारित दर (TOD)	Different rates at different hours
Power cut / outage	बिजली कटौती	Grid supply interruption

06 Money, Schemes & Buying · पैसा, योजनाएँ और खरीद

English	हिंदी	Simple meaning
Subsidy	सब्सिडी / अनुदान	Government financial support
Central financial assistance	केंद्रीय वित्तीय सहायता (CFA)	Subsidy from the central government
Eligibility	पात्रता	Conditions to qualify for a scheme
National portal	राष्ट्रीय पोर्टल	Government website for applications
Registered vendor	पंजीकृत विक्रेता	Installer approved under a scheme
Quotation	कोटेशन / मूल्य-पत्र	Written price offer
Bill of materials (BOM)	सामग्री सूची	List of all items supplied
Scope of work	कार्य का दायरा	What the installer will and won't do
Advance payment	अग्रिम भुगतान	Money paid before work
Payback period	लागत वसूली अवधि	Years for savings to equal net cost
Return on investment	निवेश पर प्रतिफल (ROI)	Gain compared with money invested
EMI / loan	ईएमआई / ऋण	Monthly loan repayment
Warranty	वारंटी	Maker's or installer's promise to repair/replace
Product warranty	उत्पाद वारंटी	Cover against manufacturing defects
Performance warranty	प्रदर्शन वारंटी	Promise of minimum output over years
Workmanship warranty	कारीगरी वारंटी	Installer's cover on installation quality
Invoice	बीजक / इनवॉइस	Final bill for the purchase
Commissioning	चालू करना (कमीशनिंग)	Final testing and start of operation

07 Farm Solar · खेती के लिए सौर

English	हिंदी	Simple meaning
Solar water pump	सोलर पंप / सौर जल पंप	Pump run by solar panels
Submersible pump	सबमर्सिबल पंप	Pump placed inside the well or borewell
Surface pump	सतही पंप	Pump placed above ground
Borewell	बोरवेल / नलकूप	Deep narrow well
Head	हेड / उठान ऊँचाई	Height water is lifted, plus pipe losses
Flow rate	प्रवाह दर	Water pumped per hour or per day

Irrigation	सिंचाई	Watering crops
Drip irrigation	टपक सिंचाई (ड्रिप)	Water delivered drop by drop to roots
Diesel pump	डीज़ल पंप	Pump run on diesel engine
Agricultural feeder	कृषि फीडर	Power line serving farm connections
Farmer producer organisation	किसान उत्पादक संगठन (FPO)	Farmers' collective enterprise
Barren / fallow land	बंजर / परती भूमि	Land not under cultivation

08 Installation, Safety & Care · स्थापना, सुरक्षा और देखभाल

English	हिंदी	Simple meaning
Installation	स्थापना / इंस्टॉलेशन	Fitting the system
Site survey	स्थल सर्वेक्षण	Inspection of roof or land before design
Waterproofing	वॉटरप्रूफिंग / जलरोधन	Keeping roof leak-free after drilling
Inspection	निरीक्षण	Check by DISCOM or authority
Electric shock	बिजली का झटका	Current passing through the body — dangerous
Arc	आर्क / चिंगारी	Electric spark across a gap; fire risk
Isolator	आइसोलेटर	Switch to disconnect a circuit for safe work
Qualified electrician	योग्य / लाइसेंसधारी इलेक्ट्रीशियन	Trained, authorised person for electrical work
Cleaning	सफ़ाई	Removing dust from panels safely
Monitoring	निगरानी (मॉनिटरिंग)	Tracking generation in an app or display
Fault / error code	खराबी / एरर कोड	Inverter warning of a problem
Hot spot	हॉट स्पॉट	Overheated area on a panel
Maintenance (O&M)	रखरखाव (ओ एंड एम)	Operation and maintenance over the system's life

09 Abbreviations · संक्षिप्त रूप

Abbreviation	हिंदी	Full form
ALMM	अनुमोदित मॉडल सूची	Approved List of Models and Manufacturers
BEE	ऊर्जा दक्षता ब्यूरो	Bureau of Energy Efficiency
BIS	भारतीय मानक ब्यूरो	Bureau of Indian Standards

CEA	केंद्रीय विद्युत प्राधिकरण	Central Electricity Authority
CERC / SERC	केंद्रीय / राज्य विद्युत नियामक आयोग	Central / State Electricity Regulatory Commission
CGRF	उपभोक्ता शिकायत निवारण फोरम	Consumer Grievance Redressal Forum
DCR	घरेलू सामग्री आवश्यकता	Domestic Content Requirement
EPC	इंजीनियरिंग, खरीद, निर्माण	Engineering, Procurement and Construction
MNRE	नवीन और नवीकरणीय ऊर्जा मंत्रालय	Ministry of New and Renewable Energy
O&M	संचालन और रखरखाव	Operation and Maintenance
PM-KUSUM	पीएम-कुसुम	Central farm solar programme
PM Surya Ghar	पीएम सूर्य घर: मुफ्त बिजली योजना	Central residential rooftop solar scheme
PV	फोटोवोल्टिक	Photovoltaic — light to electricity
RESCO	रेस्को मॉडल	Renewable Energy Service Company (pay-per-unit model)
SNA	राज्य नोडल एजेंसी	State Nodal Agency for renewable energy

10 Unit Cheat Sheet · इकाइयाँ एक नज़र में

Unit	Measures	Remember
V	Voltage	Push · India socket ≈ 230 V AC
A	Current	Flow · sizes cables and breakers
W, kW	Power	How fast · 1 kW = 1,000 W
kWh (unit)	Energy	How much · what you pay for
Wp, kWp	Panel rating	Lab (STC) power, not real-time output
VA, kVA	Apparent power	W = VA × PF
Ah	Battery charge	Wh = V × Ah
Hz	Frequency	India 50 Hz
W/m²	Irradiance	Bright noon sun ≈ 1,000 W/m²
kWh/m²/day	Irradiation	= peak sun hours · India mostly 4–7 (MNRE)
HP	Motor power	1 HP ≈ 0.746 kW
Ton (AC)	Cooling capacity	1 ton ≈ 3.5 kW of cooling — not electricity used

Son of Surya

LEARN WITH SON OF SURYA

Every term here is explained in depth across the free foundation guides R1, R2, R3 and SOS-01. A full Hindi edition of each guide is in preparation.

11 Sources & Disclaimer

Definitions are simplified from standard electrical and photovoltaic terminology and from the terms used by MNRE, CEA, BIS and Indian DISCOMs. Hindi equivalents reflect common Indian usage and standard technical Hindi; formal government glossaries may use different words. Scheme names are given for recognition only — check current details on official portals. This dictionary is for general education and is not a legal or technical standard. Published by Son of Surya, October 2026.

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FREE FOUNDATION GUIDES

- **R1** Electricity Basics for Solar
- **R2** How to Read Your Electricity Bill
- **R3** Sun, Site & Shade
- **SOS-01** Solar Starter Guide
- **R11** Solar Dictionary (English–Hindi)

SPECIALISED GUIDES — RELEASING IN BATCHES

- **Home** — rooftop planning, sizing, net metering
- **Kisan** — farm solar, pumps, schemes
- **Business & institutions** — shops, schools, societies
- **Professional** — wiring, installation, EPC

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